Kashiful Haque

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ML Engineer with 4 YOE across backend infrastructure, LLM pre-training, post-training with RL, and building inference engines in C++ and Rust. Pre-trained a 360M LLaMA from scratch, fine-tuned production LLMs, and applied GRPO/ORPO for behavioral alignment. Looking for roles in pre-training, RL-based post-training, and GPU inference — CUDA, HPC, and low-level runtime work.

Work Experience

wand.ai
Bangalore

Backend AI / ML Engineer — 11/2025 – Present

- Built observability for agent workflows: span hierarchy, trace IDs, and Kafka-based monitoring events for e2e run visibility.
- Implemented OAuth/OIDC credential flows with deferred auth, token refresh, and variable propagation across agentic tasks.
- Built workflow specs: generates auth gateways, OAuth triggers, and injects credential references into tool environments.
- Added dynamic tool support: agents can define, reference, and execute tools that resolve workflow state variables at runtime.

American Express
Bangalore

Engineer III — 02/2025 – 11/2025

- Built a hybrid search over 200k+ internal docs combining dense embeddings and keyword retrieval; cut p95 query latency from 30s to 2s via caching and query optimizations.
- Ran large-scale transformer training and profiling experiments (FlashAttention, KV-cache, batching) to improve embedding throughput and serving efficiency.

Fiery
Bangalore

Associate Software Engineer — 01/2023 – 02/2025

- Fine-tuned domain LLMs using SFT/QLoRA, served via vLLM on T4 clusters with sub-second p95 TTFT.
- Built Fiery Scribe: translates print requests to printer XML using ModernBERT + LLM.
- Built AskDB: an LLM agent that maps natural language to SQL over production DBs and generates business reports.

ML Engineering Work

banana.cpp: LLM Inference Engine in C++ • github
- Inference engine with KV-cache, speculative decoding, and continuous batching; 10x speedup via CPU parallelization and fused kernel optimizations.
- Focused on Apple M-series silicon as a learning project into low-level inference mechanics.

Llama-3.2 3B Politeness Alignment using ORPO • hf • blog
- Applied ORPO with TRL + Unsloth to align Llama-3.2 3B to a behavioral policy (respond politely or refuse).
- Built a 32k-row preference dataset with behavioral constraints; validated convergence via reward margins and log-prob dynamics in W&B.

smol-llama: 360M LLM pre-training • github • hf • wandb • blog
- Trained a 360M LLaMA from scratch in PyTorch with GQA, RoPE, RMSNorm, and SwiGLU.
- Pre-trained on 6B tokens using FlashAttention on H100 at 75K tokens/sec; built a cost-efficient pipeline with gradient accumulation and checkpointing.

smoltorch: minimal autograd engine • github • pypi • blog
- Autograd engine with computation graphs and topological scheduling, built to study gradient propagation.

Education

IIT Madras
BS, Data Science and Applications — 2020 – 2024

Skills
- Python, C++, Rust, Go
- PyTorch, FlashAttention, CUDA, vLLM, TRL, QLoRA, Unsloth, Docker, Kubernetes, Redis, PostgreSQL